Feature Speed, Transaction Costs, and the Limits of Machine Learning in Emerging Markets

Kanpat Vesessook*
SCBX
Bangkok, Thailand
Brown University
Providence, RI, USA
kanpat_vesessook@brown.edu

Nut Chukampaeng
SCBX
Bangkok, Thailand
nut.c@scbx.com

Abstract

Across 15 emerging markets, eight portfolio interventions converge on the same net Sharpe ceiling (~0.45–0.49 at 3× transaction costs), regardless of whether they restrict features, smooth predictions, or reduce rebalancing frequency. The ceiling follows a cost-speed frontier: $\mathrm{SR}_{\mathrm{gross}} - \mathrm{SR}_{\mathrm{net}} \approx \lambda\kappa \cdot \mathrm{FSI} + c$, where the Feature Speed Index (FSI) measures a strategy's ex-ante exposure to fast-moving characteristics. FSI explains 64–65% of cross-sectional SR degradation at each cost level ($\lambda = 0.33$ per unit κ; pooled $R^2 = 0.72$ across 300 strategy–market–cost combinations) and predicts realized turnover ex ante ($R^2 = 0.73$). Non-smooth models (trees, neural networks) fall systematically below the frontier, so model smoothness is a second constraint. The frontier explains why ML's prediction advantage (gross SR 0.68 vs. sorts' 0.58) vanishes after costs: ML loads on fast features that push its turnover above that of sorts (23–38% vs. 17%). Restricting Ridge regression to slow, persistent characteristics is the simplest path to the ceiling (SR 0.47, 10% turnover); EMA smoothing and quarterly rebalancing reach the same ceiling through different mechanisms, while cost-aware neural networks give up too much signal to cut turnover and fall short.

CCS Concepts

• **Applied computing** → **Economics**; • **Computing methodologies** → **Machine learning**.

Keywords

machine learning, emerging markets, transaction costs, portfolio construction, factor investing

ACM Reference Format:
Kanpat Vesessook and Nut Chukampaeng. 2026. Feature Speed, Transaction Costs, and the Limits of Machine Learning in Emerging Markets. In *Proceedings of the 7th ACM International Conference on AI in Finance (ICAIF '26), November 14–17, 2026, Milan, Italy*. ACM, New York, NY, USA, 8 pages.

*Work done during an internship at SCBX.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
ICAIF '26, Milan, Italy
© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

1 Introduction

Machine learning has transformed cross-sectional asset pricing. Gu, Kelly, and Xiu [8] show that tree-based and neural network models produce superior out-of-sample R^2 for US equities, and Hanauer and Kalsbach [9] extend this to emerging markets (EM). More recently, Jensen, Kelly, Malamud, and Pedersen [10] (JKMP) identify that cost-agnostic ML overweights "fleeting small-scale characteristics," proposing an "implementable efficient frontier" that integrates costs into the optimization objective.

We build on their diagnosis and show that the cost–speed interaction creates a *performance ceiling*. Across 15 emerging markets, eight intervention variants from four mechanism families (feature selection, persistence filtering, prediction smoothing, quarterly rebalancing) all reach net SR ~0.45–0.49 at 3× EM costs. JKMP optimize toward an implementable frontier; we characterize the ceiling that any method, upstream or downstream, approaches. Our main contribution is an ex-ante diagnostic for that ceiling, the Feature Speed Index (FSI). Computed from training data alone, FSI explains 64–65% of cross-sectional cost-driven SR degradation at every cost level and 73% of realized turnover.

We use 154 firm-level characteristics from the JKP Global Factor Data [11], walk-forward validation, and Corwin-Schultz [3] transaction costs. The analysis yields five findings:

(1) **A cost-speed ceiling.** Eight intervention variants converge on net SR ~0.45–0.49 despite different mechanisms. The frontier $\mathrm{SR}_{\mathrm{gross}} - \mathrm{SR}_{\mathrm{net}} \approx \lambda\kappa \cdot \mathrm{FSI} + c$ fits SR degradation with pooled $R^2 = 0.72$ across 300 strategy–market–cost combinations; at each fixed cost level, FSI explains 64–65% of cross-sectional degradation ($\lambda = 0.33$ per unit κ).
(2) **Cost–speed interaction.** At zero costs, fast and slow features produce comparable gross SRs (0.63 vs. 0.61). At 3× EM costs, fast features degrade ~3× faster (SR degradation 0.42 vs. 0.14).
(3) **Simple upstream fixes reach the ceiling.** Restricting Ridge to slow features (57 characteristics) achieves net SR 0.47 at 10% turnover vs. standard Ridge (0.35, 23%) and MomentumSort (0.36, 17%), outperforming the sort in 11/15 markets (not individually significant; Section 6).
(4) **ML overweights fast features.** ML allocates 43–47% of importance to fast features (25% of the set), driving 63% of prediction changes.
(5) **Prediction stability is the binding constraint.** Tree models pair the strongest gross signals with the worst net SRs (XGBoost 0.66 gross, 0.17 net; Table 1) and slow features do

not help. Walk-forward EMA smoothing lifts XGBoost only to sort parity (0.35), short of the frontier.

On US equities (top 500, 2021–2025), the decomposition holds: Ridge_Slow beats Ridge_Fast at every cost level, and Ridge_Fast collapses at 3× costs (SR −0.19 vs. 0.51). Our results add a simpler upstream alternative to JKMP's downstream approach and give large-scale EM evidence for Gârleanu and Pedersen's [7] prediction that optimal portfolios overweight slow-decaying signals.

2 Related Work

Cost-aware ML.. JKMP [10] integrate transaction costs into the ML portfolio optimization objective. We differ in three respects: (i) a quantitative feature-level diagnostic (rank autocorrelation classifies which features drive turnover), (ii) demonstration that the problem is cost-regime dependent (modest in US, catastrophic in EM), and (iii) a simpler upstream fix (slow-feature selection, net SR 0.47 vs. MomSort 0.36) that avoids cost-aware optimization entirely.

ML for asset pricing. Gu et al. [8] establish ML's superiority for US equity prediction. Hanauer and Kalsbach [9] extend this to EM, finding ML generates higher long-short Sharpe ratios than linear baselines. We confirm the ML-vs-linear ranking but show that after costs in long-only portfolios, sorts match or beat ML. Esakia and Goltz [5] reach a similar conclusion for long-only settings. Azevedo et al. [1] find 120–140% monthly turnover and 13–40% performance reduction from costs in ML strategies. Chen et al. [2] document that researcher degrees of freedom produce non-standard errors ∼1.59× the standard error.

Optimal trading with slow signals. Gârleanu and Pedersen [7] (GP) prove that under transaction costs, optimal portfolios overweight slow-decaying signals. Our results validate GP's predictions in a setting they did not study: ML portfolios with 154 characteristics across 15 EM markets.

Transaction costs and implementability. Realistic trading costs render many anomalies unprofitable [13], favor diversified factor combinations [4], and reward low-turnover construction [6], including turnover-aware EM factor designs [14].

3 Data and Methodology

3.1 Data

We use monthly equity data for 15 emerging markets from the JKP Global Factor Data [11]: BRA, CHN, GRC, IDN, IND, KOR, MEX, MYS, PHL, POL, SAU, THA, TUR, TWN, and ZAF. Each provides 154 firm-level characteristics. The prediction target is the one-month excess return (`ret_1_0`). For each market-month, we restrict to the top N stocks by market cap ($N \in [50, 300]$; most at 100). Features are lagged *before* universe filtering to avoid collider bias, since $\text{me}(T) = \text{me}(T-1) \times (1+r_T)$.

3.2 Strategies

We compare seven strategies: three factor sorts (MomentumSort: 12-month return with 1-month skip; LowVolSort: 60-day volatility; CompositeSort: z-score of momentum, value, and quality), three ML models (Ridge α=100; XGBoost depth 4, 200 trees; LightGBM 31 leaves, 200 trees), and a MomentumML_Blend (80% momentum + 20% Ridge rank). All use top-quintile, long-only, equal-weighted portfolios with a 20% rebalancing buffer. All ML models train on cross-sectional rank targets: $\tilde{y}_{i,t} = \text{Rank}(r_{i,t})/N_t$.

3.3 Walk-Forward Validation

We use expanding-window walk-forward with 60-month minimum training. Models retrain monthly on all available data up to time t. The OOS period begins 2016–2018 (market-dependent), yielding 56–93 OOS months.

3.4 Transaction Cost Model

Costs use the Corwin-Schultz [3] half-spread with multiplier κ: $c_{i,t} = \kappa \times \text{HalfSpread}_{i,t}$, clipped to $[20, 200]$ bps. Our baseline is $\kappa = 3$, following Novy-Marx and Velikov [13] and validated against actual costs from a mid-size EM equity fund (∼70–100 bps one-way). Median CS half-spreads on our EM universes are 30–40 bps versus ∼25 bps for US large-caps; the EM cost gap is driven mainly by market impact from thinner order books, captured by κ. For robustness, we also test flat cost schedules (10–100 bps per trade).

3.5 Performance Metrics

Sharpe ratios are computed as $\text{SR} = (\bar{r}_p/\sigma_p) \times \sqrt{12}$, where r_p is the monthly net-of-cost portfolio return. Since all strategies are long-only in the same cap-filtered universe, relative comparisons are unaffected by the common market-beta component. We additionally report an *alpha Sharpe ratio* $\text{SR}_\alpha = \text{SR}(r_p - r_{\text{EW}})$ that isolates stock-selection skill by subtracting the equal-weight universe return each month.

3.6 The Feature Speed Index and the Cost-Speed Frontier

To quantify ex-ante exposure to prediction instability, we define the Feature Speed Index (FSI) as the equal-weight average instability of a strategy's feature set S:

$$\text{FSI}(S) = \frac{1}{|S|} \sum_{j \in S} (1 - \text{AC}_j), \tag{1}$$

where AC_j is the cross-sectional Spearman rank autocorrelation of feature j, computed on a rolling 60-month training window. This feature-set definition is available before model fitting for all strategies and therefore avoids using realized model coefficients in the diagnostic. For single-feature sorts, S contains only the sorting characteristic.

The FSI enters a linear frontier for cost-driven performance loss:

$$\text{SR}_{\text{gross}} - \text{SR}_{\text{net}} \approx \lambda\kappa \cdot \text{FSI} + c, \tag{2}$$

where κ is the cost multiplier, λ captures the sensitivity of degradation to feature speed, and c is the baseline cost drag. Under proportional costs, κ-scaling is mechanical; the frontier's informative content is *cross-sectional* (Section 4.6). It is estimated on smooth models (Ridge, sorts) where prediction surfaces are continuous; non-smooth models exhibit systematic excess degradation beyond it (Section 4.5).

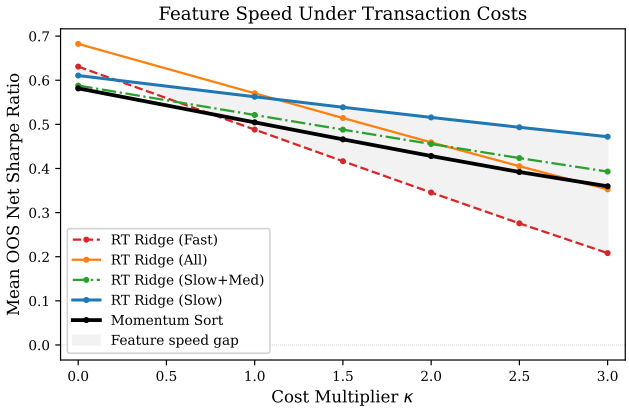


Figure 1: Net SR vs. cost multiplier κ. Fast features degrade ~3× faster than slow features. The shaded region shows the feature-speed gap, which widens monotonically with κ. At zero costs, fast and slow are competitive; at 3× EM costs, the gap exceeds 0.25 SR.

4 Results

4.1 The Cost–Speed Interaction

We classify 154 features into three speed categories: **SLOW** (57 accounting fundamentals, rank AC 0.97), **MEDIUM** (56 intermediate features, AC 0.83), and **FAST** (38 market microstructure variables, AC 0.68). The three classes differ significantly in rank AC (Kruskal-Wallis $p = 3.77 \times 10^{-11}$).

Figure 1 plots net Sharpe ratio against the cost multiplier κ. At zero costs, fast and slow features give similar gross SRs (0.63 vs. 0.61), so fast features do carry signal. As costs rise, fast-feature performance degrades ~3× faster (degradation 0.42 vs. 0.14 SR at 3×), because the cost of their extra turnover grows linearly with κ. Fast features fail because of what they cost to trade.

In EM, the cost–speed interaction is 3–5× larger than in US markets, suggesting that downstream optimization alone may not suffice; the problem must also be addressed upstream.

4.2 ML vs. Sorts: Ridge Matches Momentum

With rank-target training, Ridge essentially matches MomentumSort across 15 EM markets at 3× costs: average net SRs are Blend 0.38, MomSort 0.36, Ridge 0.35, CompositeSort 0.32, XGBoost 0.17, LightGBM 0.13. Ridge wins 8/15 markets (Wilcoxon $p = 0.45$, not significant); tree models are significantly worse ($p < 0.01$).

Table 1 isolates the cost interaction. Ridge generates a *stronger* gross signal (0.68) than MomSort (0.58), but 23% turnover vs. 17% erases the advantage.

Per market (Table 2), Ridge leads where its turnover is contained (TWN: 0.73 vs. 0.51; CHN: 0.23 vs. 0.07). Ridge_Slow (Section 4.4) is best in 5 markets, positive in 13/15.

4.3 The Feature-Speed Decomposition

We now trace why ML loads on fast features and how that loading becomes portfolio turnover. FAST features make up 25% of the feature set but receive 43% of model importance in Ridge and 47% in XGBoost, and they generate 63% of month-to-month prediction changes (Table 3). Their low rank autocorrelation (0.68 vs. 0.97 for SLOW) passes directly into the predictions.

Table 1: Gross vs. net Sharpe ratios at 3× costs, averaged across 15 EM markets. Ridge has a stronger gross signal but higher turnover erases the advantage.

Strategy	Gross SR	Net SR	Turnover	Cost Drag
MomentumSort	0.58	0.36	17.0%	~590 bps
Ridge	0.68	0.35	23.3%	~700 bps
XGBoost	0.66	0.17	36.3%	~1,100 bps
LightGBM	0.64	0.13	38.2%	~1,160 bps
Blend (80/20)	0.63	0.38	19.0%	~640 bps

Stock-level IC.. Ridge_Slow has the highest IC×stability (0.030; Table 4): its IC matches momentum's (0.031) and its rank stability is 0.939. Ridge_Fast has the lowest stability (0.539) and no offsetting gain in IC.

Model-free signal decay. The speed differential is a data property, not a modeling artifact. With no fitted model, we compute per-feature monthly rank ICs against single-month returns at horizons of 1–12 months, sign-aligned per feature. One-month ICs are comparable (slow 0.027, fast 0.025), but slow features retain 69% of their IC at a 12-month horizon versus 57% for fast features (Wilcoxon $p = 0.01$ across 15 markets; Figure 2). Fast features' predictive *content* decays only moderately faster; what churns is their *values* (rank AC 0.68 vs. 0.97, Table 3), and that churn forces turnover. As in Section 4.1, fast signals stay informative but are expensive to trade.

Summary. FAST features $\xrightarrow{\text{high importance}}$ unstable predictions $\xrightarrow{\text{high turnover}}$ low net SR.

4.4 Upstream Fix: Slow-Feature Ridge

The decomposition suggests a simple upstream fix: train only on slow, persistent characteristics. Ridge_Slow reaches net SR 0.47 at 10% turnover, against 0.35 for Ridge_All and 0.36 for MomSort (Table 5). Ridge_Slow uses the same model and hyperparameters as Ridge_All, so the gain comes entirely from feature selection. It beats MomSort in 11/15 markets, though the difference is not significant (pooled block bootstrap $p = 0.10$; Section 6).

Aggregation versus speed. Comparing Ridge and CompositeSort at matched dimensionality ($N \in \{5, 10, 20, 50\}$): at $N = 5$, turnover is nearly identical (5.1% vs. 5.6%), ruling out regression itself as the turnover source.

Data-driven persistence filter. A persistence filter (rank AC > 0.75, recomputed on training data; ~120 features) and a speed-tercile split (~127 features) both yield net SR 0.47 at 13% turnover, matching the hand-classified slow set while keeping about twice as many features.

The fix needs no cost-aware optimization, prediction smoothing, or change to portfolio construction.

Table 2: Out-of-sample net Sharpe ratios at 3× costs across 15 emerging markets. N is the universe size (top stocks by market cap); T_{OOS} is the number of out-of-sample months. Bold indicates the best active strategy per market. EW is the equal-weighted passive benchmark. Ridge matches MomSort on average (0.35 vs. 0.36) but generates higher turnover (23% vs 17%); Ridge_Slow (Section 4.4) reaches 0.47 at 10% turnover.

Market	N	T_{OOS}	EW	MomSort	Composite	Ridge	Ridge_Slow	XGBoost	LightGBM	Blend	TO_{Mom}	TO_{Ridge}
BRA	100	83	0.10	**0.08**	−0.04	−0.05	−0.06	−0.30	−0.23	**0.08**	18%	30%
CHN	300	78	0.24	0.07	0.13	0.23	**0.39**	−0.07	−0.17	0.08	21%	23%
GRC	50	78	0.64	0.43	**0.92**	0.73	0.74	0.64	0.50	0.49	15%	19%
IDN	100	80	0.29	**0.59**	0.13	0.42	0.54	0.22	0.14	0.47	17%	18%
IND	200	81	0.61	**0.63**	0.29	0.37	0.58	0.09	−0.11	0.59	16%	28%
KOR	200	86	0.07	0.03	−0.32	**0.07**	0.07	−0.23	−0.29	−0.04	18%	15%
MEX	100	93	0.46	0.25	0.40	0.18	**0.51**	−0.24	−0.32	0.23	18%	27%
MYS	100	90	0.23	0.39	0.14	**0.46**	0.43	0.24	0.28	**0.46**	15%	22%
PHL	100	87	−0.02	0.01	**0.41**	0.10	0.16	−0.14	−0.11	0.17	15%	25%
POL	100	69	0.53	0.55	0.60	0.57	**0.70**	0.60	0.57	0.55	17%	24%
SAU	100	56	1.01	1.13	0.93	0.85	1.20	0.93	0.94	**1.22**	14%	25%
THA	100	84	−0.09	0.01	−0.18	−0.05	−0.06	−0.38	−0.43	**0.09**	17%	20%
TUR	100	79	0.61	0.33	**0.67**	0.45	0.54	0.39	0.43	0.35	18%	24%
TWN	200	78	0.67	0.51	0.54	0.73	**0.86**	0.71	0.73	0.51	17%	23%
ZAF	100	93	0.30	0.39	0.24	0.24	**0.49**	0.07	−0.02	0.48	18%	28%
Mean	–	80	0.38	0.36	0.32	0.35	**0.47**	0.17	0.13	0.38	17%	23%

Table 3: Feature speed decomposition. FAST features dominate both model importance and prediction changes despite representing a minority of the feature set.

Speed	# Feat.	Rank AC	% Import.	% Pred. Δ
SLOW	57	0.97	31–34%	12.6%
MEDIUM	56	0.83	22–23%	24.1%
FAST	38	0.68	43–47%	63.4%

Table 4: Information coefficient analysis across 15 EM markets. IC×Stability captures the effective signal after accounting for prediction churn.

Strategy	IC	Stability	IC×Stab	Hit Rate
MomentumSort	0.031	0.894	0.028	57.6%
Ridge_All	0.022	0.665	0.015	57.0%
Ridge_Slow	0.031	0.939	**0.030**	57.7%
Ridge_Fast	0.013	0.539	0.007	54.0%

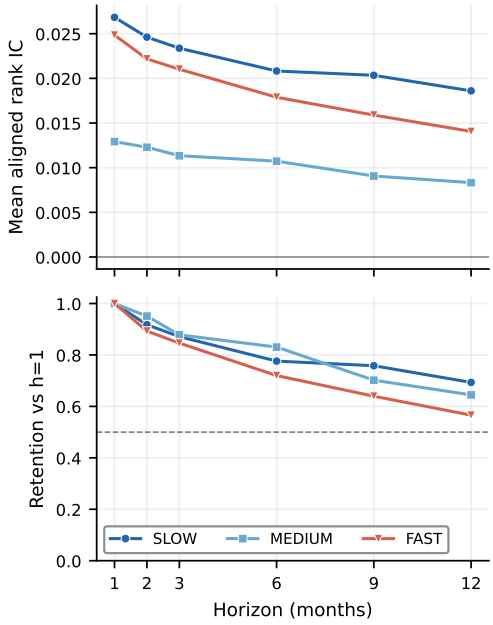


Figure 2: Model-free rank-IC decay by feature speed class. Slow features retain 69% of their one-month IC at 12 months versus 57% for fast features; no model is fitted anywhere in this analysis.

Alpha over the passive benchmark. To isolate stock-selection skill, we compute $SR_\alpha = SR(r_p - r_{EW})$, where r_{EW} is the equal-weighted universe return. At 3× costs, Ridge_Slow has $SR_\alpha = +0.18$, positive in 10/15 markets, and MomSort +0.07 (8/15). Ridge_All (−0.15, 6/15) and Ridge_Fast (−0.46, 2/15) are negative. At 1× costs, all variants produce comparable positive alpha, so the gap comes from costs.

Table 5: Feature-speed ablation: Ridge with different feature subsets. Mean OOS net SRs across 15 markets at 3× costs.

Strategy	Net SR	Turnover	# Features
Ridge_Slow	**0.47**	**10.2%**	57
Ridge_SlowMed	0.39	14.5%	113
MomentumSort	0.36	17.0%	1
Ridge_All	0.35	23.3%	151[†]
Ridge_Fast	0.21	30.1%	38

[†]154 characteristics minus me (universe filter), target, and identifier.

Table 6: The prediction-portfolio disconnect across model classes and smoothing interventions. Sorted by gross SR. All rows use the tree-comparison sample: walk-forward over all months after the initial 60-month training window (∼102 months per market), a longer window than Tables 1–2's OOS period, so levels are not comparable across tables.

Model	Strategy	Gross SR	Net SR	SR Lost	TO
XGB	All	0.767	0.300	−0.467	34.5%
LGBM	All	0.748	0.265	−0.482	35.5%
Ridge	All	0.684	0.357	−0.326	23.0%
XGB	Slow	0.647	0.263	−0.385	28.6%
Ridge	Slow	0.616	0.479	−0.137	9.9%
MomSort	—	0.606	0.373	−0.233	17.2%

4.5 Model Complexity Amplifies the Cost Channel

Tree-based models expose a second mechanism, model complexity, that widens the gap between gross and net performance independently of feature speed. On a common sample (Table 6), XGBoost has the highest unsmoothed gross SR (0.77), yet its net SR (0.30) falls well below Ridge_All (0.36) and Ridge_Slow (0.48). Gross and net rankings are negatively correlated (rank correlation −0.52 across model–feature combinations).

Slow features do not help trees; smoothing only partly does. XGB_Slow (net SR 0.263) does *worse* than XGB_All (0.300): shifting decision boundaries create turnover even when the inputs are stable. Output smoothing helps more, but only partly. On the main OOS sample, walk-forward EMA lifts XGBoost from 0.24 to 0.35 and cuts turnover from 36% to 25%, which reaches sort parity but not the frontier. The best fixed α (0.2, chosen post hoc) reaches 0.45 at 12% turnover, still below Ridge_Slow's 0.47. An earlier post-2020 validation had suggested that smoothed trees beat Ridge_Slow (0.53); this does not replicate on the full sample.

A 2 × 2 framework. The results decompose into feature speed × model smoothness (Figure 3): Ridge_Slow loses only 0.14 SR to costs, Ridge_All 0.33, XGB_Slow 0.39, XGB_All 0.47. Output smoothing partially bridges the gap (walk-forward EMA cuts XGBoost's gross-to-net loss from 0.49 to 0.33 on the main OOS sample), which points to prediction instability rather than model capacity as the mechanism.

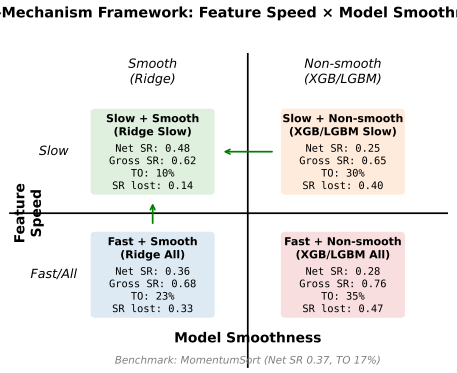


Figure 3: The 2×2 framework: feature speed × model smoothness. Both slow features and a smooth model are needed to minimize the prediction-portfolio disconnect.

Implication. Net performance is limited by prediction *stability* more than accuracy. EMA improves both Ridge (+0.10 SR) and XGBoost (+0.11 walk-forward, +0.21 post hoc) but cannot fully offset piecewise-constant surfaces. Residual instability absorbs trees' stronger gross signal (0.72 unsmoothed vs. 0.61 for Ridge_Slow on the main OOS sample), which is why simple linear models remain competitive after costs [5, 8].

4.6 The Cost-Speed Frontier

We estimate the frontier (Equation 2) on smooth models (Ridge variants and sorts) over 300 strategy–market–cost combinations: 5 strategies, 15 markets, and 4 cost levels. Figure 4 plots SR degradation against $\kappa \cdot \text{FSI}$, with pooled $R^2 = 0.72$. Part of this fit is mechanical. Costs are proportional to κ, so the slope scaling with κ shows only that these strategies trade the same way at every cost level. The informative result is cross-sectional. At each fixed κ, FSI alone explains 64–65% of degradation across the 75 strategy–market cells; κ alone explains 37% of the pooled variation.

We report the per-κ slope, $\lambda = 0.33$, throughout. Its cluster-robust 95% CIs are $[0.27, 0.40]$ by market and $[0.26, 0.41]$ by strategy. The pooled slope (0.39) gives more weight to high-κ cells. FSI also predicts turnover, explaining 73% of realized turnover across cells, and it is computed from training data *before* any trading. It works at the strategy level. FSI is a property of the feature set and barely varies across markets within a strategy, so it separates feature sets and not markets (within-strategy $R^2 = 0.07$). Its coefficient does not change when we control for each market's spread level.

We then add the tree models (XGBoost, LightGBM; same OOS window as Table 1) with an indicator for non-smooth models, which tests the smoothness channel directly. At $\kappa = 3$, the fit on this pooled sample rises from $R^2 = 0.34$ without the indicator to 0.73 with it. Trees lose an extra 0.22 SR, and the FSI coefficient does not change. EMA smoothing moves trees toward the frontier without reaching it (Section 4.5).

In Figure 5, at 3× EM costs, eight strategies cluster at net SR 0.45–0.49 despite different mechanisms: slow-feature selection (Ridge_Slow 0.47), persistence filtering (0.47), speed-tercile selection (0.47), a data-driven rank-target variant (0.46), fixed and

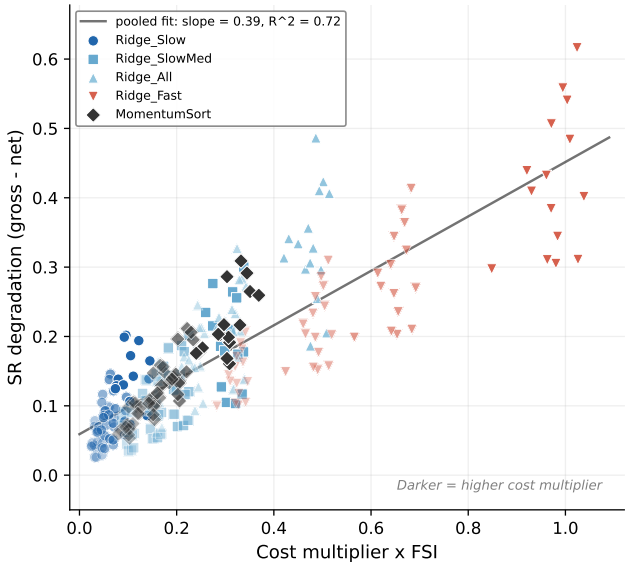


Figure 4: The cost-speed frontier (Equation 2): SR degradation vs. $\kappa \times$ FSI across 300 strategy–market–cost combinations. Pooled $R^2 = 0.72$; cross-sectional $R^2 = 0.64$–0.65 at each fixed κ with slope $\lambda = 0.33$ per unit κ. Darker points indicate higher κ.

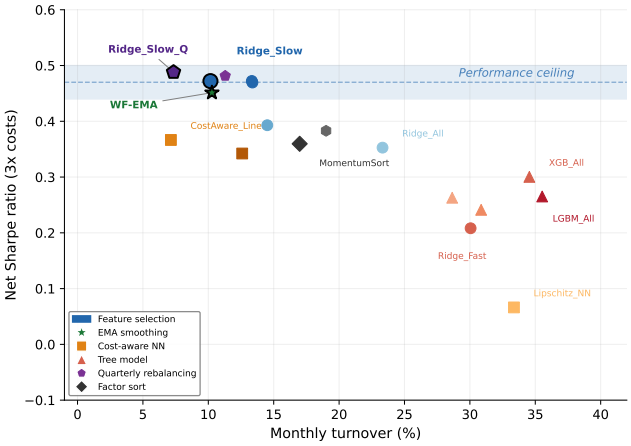


Figure 5: The cost-speed frontier at 3× EM costs. Strategies that minimize FSI converge on net SR ∼0.45–0.49 (shaded band). Below the ceiling, net SR degrades with turnover along the frontier. Color indicates mechanism type.

walk-forward EMA smoothing (0.47 and 0.45), and quarterly rebalancing (Ridge_All_Q 0.48, Ridge_Slow_Q 0.49). Strategies that fail to minimize FSI (tree models, fast-feature Ridge, unconstrained NNs) fall below the ceiling in proportion to $\kappa \cdot$ FSI.

The ceiling height depends on $\mathrm{SR}_{\mathrm{gross}}$ and the market's cost environment; the frontier's portable content is the cross-sectional FSI–degradation relationship at each cost level.

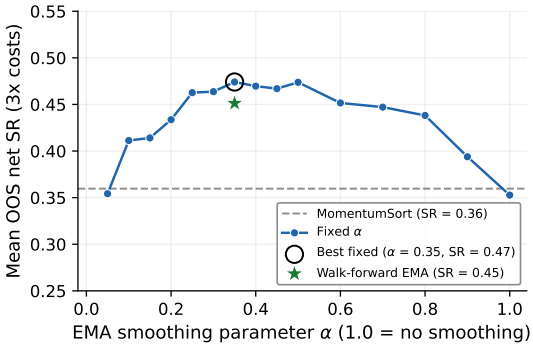


Figure 6: EMA smoothing: α vs. net SR at 3× costs (15 markets). Walk-forward EMA (green star) achieves 0.45, beating MomSort (0.36).

Table 7: Prediction smoothing results (15 markets). WF-EMA selects α inside the walk-forward loop using recent in-sample months and does not use future OOS months.

Strategy	Net SR	Gross SR	TO	Stability	Wins
Ridge (no smooth)	0.35	0.68	23.3%	0.750	8/15
MomentumSort	0.36	0.58	17.0%	—	—
Ridge WF-EMA	**0.45**	**0.60**	**10.3%**	**0.919**	**10/15**

Cross-market transfer. To test whether the frontier generalizes out of sample, we hold out markets. Leave-one-market-out (15 folds) gives mean OOS $R^2 = 0.60$, and leave-5-out (3,003 splits) gives 0.69, with 97% of splits above 0.50. The pooled slope barely moves across folds (range $[0.37, 0.40]$; bootstrap 95% CI $[0.34, 0.45]$), so no single market drives the relationship. The slow-feature set transfers too: 52 of the 57 slow features are slow in *all* 15 markets (mean Jaccard similarity 0.97), so "slow" can be defined on one group of markets and applied to another.

4.7 Prediction Smoothing

A complementary intervention smooths predictions at the output level. We apply an EMA to Ridge's prediction ranks: $\hat{r}_{i,t}^{\mathrm{smooth}} = \alpha \cdot \hat{r}_{i,t}^{\mathrm{raw}} + (1-\alpha) \cdot \hat{r}_{i,t-1}^{\mathrm{smooth}}$, where lower α means more smoothing. Figure 6 shows the alpha–SR relationship. The curve peaks at $\alpha = 0.35$ (net SR 0.47), forming an inverted-U. Walk-forward alpha selection, maximizing IS net SR over the last 24 training months scored by the current-window model, achieves 0.45 without using future OOS months, beating MomSort (0.36) in 10/15 markets.

In Table 7, WF-EMA's gross SR (0.60) is *below* unsmoothed Ridge (0.68): the +0.10 net SR improvement comes entirely from lower turnover (23% → 10%). EMA helps Ridge but *hurts* MomSort (−0.04), confirming it recovers ML-specific signal.

EMA is redundant with slow features. Ridge_Slow with EMA selects $\alpha = 1.0$ (no smoothing): when features are already slow, predictions are inherently stable (rank stability 0.939, Table 4). The two interventions are substitutes.

Table 8: US equities: net SR by strategy and cost multiplier. Top 500, OOS 2021–2025. Ridge_Slow outperforms Ridge_Fast at every cost level; Ridge_Fast goes negative at 3×.

κ	MomSort	Ridge_All	Ridge_Slow	Ridge_Fast
0×	0.775	0.580	0.638	0.404
1×	0.693	0.445	0.574	0.206
2×	0.612	0.310	0.510	0.008
3×	0.530	0.175	0.510	−0.189

EMA helps tree models, but less reliably. For XGBoost (0.24 to 0.35; Section 4.5), the optimal smoothing is stronger ($\alpha = 0.2$ vs. 0.35), reflecting greater instability, and walk-forward selection captures barely half the post-hoc gain (+0.11 of +0.21; Ridge captures +0.10 of +0.12) because trees' noisier in-sample net SR misguides the α choice.

4.8 Cross-Regime Validation: US Equities

We test the identical pipeline on US equities (top 500 by market cap, OOS 2021–2025, 50 months). Table 8 confirms the decomposition is not EM-specific: Ridge_Slow outperforms Ridge_Fast at every cost level, and Ridge_Fast collapses at 3× costs (−0.19 vs. 0.51). MomentumSort retains a clear advantage in the US; we do not claim Ridge beats MomSort there. The EM–US divergence reflects two channels: MomSort's gross signal is substantially stronger in the US (0.78 vs. 0.58 in EM), and EM costs are 3–5× larger, making the turnover reduction from slow-feature selection more valuable relative to the sort's advantage. What replicates is the ordering of Ridge's feature subsets.

Two channels. The decomposition operates through signal degradation and cost amplification. Ridge_Fast trails Ridge_Slow even at zero costs (0.40 vs. 0.64), and the gap widens from 0.23 to 0.70 SR as κ rises from 0 to 3. In EM, ML concentrates importance on fast features well beyond their share of the feature set (Table 3), compounding both channels.

5 Discussion

Theoretical foundation: GP, JKMP, and the ceiling. Our cost-speed frontier (Equation 2) operationalizes Gârleanu and Pedersen's [7] (GP) insight that optimal portfolios overweight slow-decaying signals. GP prove this in continuous time with known dynamics; we show the same structure creates a quantifiable ceiling in a discrete-time ML pipeline with 154 estimated characteristics. Under proportional costs, a per-κ slope of exactly $\lambda\kappa$ means strategies trade identically at every cost level. This describes standard pipelines rather than a general law: a cost-aware (JKMP-style) strategy would cut turnover as κ rises and bend the frontier (λ decreasing in κ). Standard ML pipelines do not adapt this way, which motivates the upstream fix. In EM, where cost estimates are noisy, that fix reaches the ceiling without reliable cost inputs.

The 2×2 framework (Section 4.5) explains why non-smooth models stay below the ceiling: feature selection cannot fix a non-smooth model. On the same slow features, the excess degradation of trees and unconstrained NNs correlates $r = 0.91$ with model-induced turnover.

Reconciling with Hanauer and Kalsbach. Hanauer and Kalsbach [9] show ML produces superior long-short Sharpe ratios vs. linear models in EM. We confirm their core finding: ML generates stronger *gross* signals (Ridge gross SR 0.68 vs. MomSort 0.58). The divergence arises from the evaluation framework: we benchmark against factor sorts (sparse, low-turnover, and thus a harder cost-adjusted benchmark); we evaluate long-only portfolios, since short-selling is restricted in most EM (Section 6); and at realistic EM costs ($\kappa = 3$, ∼90 bps one-way), ML's turnover from fast-feature loading erases the signal advantage. Our results therefore complement theirs by locating the cost level at which ML's stronger signal stops producing better portfolios.

Practical recommendations. For EM equity portfolios: (1) without ML infrastructure, default to momentum sorts; (2) teams with ML should restrict to slow features and train on rank targets, which beats the sort without prediction smoothing or cost-aware optimization. In our data, fast features' gross advantage survives only below $\kappa \approx 0.2$ (∼6 bps one-way): at all realistic EM cost levels, slow features dominate. The FSI provides an ex-ante screen: compute Equation 1 before deployment, and if FSI exceeds ∼0.15, restrict features or apply smoothing.

Limitations. Our OOS spans 56–93 months per market with cross-country correlation ∼0.6, limiting statistical power (see Section 6 for formal tests). The Corwin-Schultz cost model with 3× multiplier is validated against an EM equity fund but remains a proxy. We test only large-caps; ML may fare better in small-caps, but costs there are higher. Tree hyperparameters are fixed at standard values [8], making our comparison conservative. Implementation choices affect results [2], though the feature-speed decomposition reflects data properties rather than modeling choices. Combining upstream feature selection with JKMP-style downstream optimization is untested.

6 Robustness and Alternatives

Cost-aware neural networks. Cost-aware training objectives could address turnover without discarding features. We test three alternatives, *all restricted to the 57 slow features*: (i) **CostAware_Linear** (linear model with differentiable net-of-cost Sharpe loss), (ii) **CostAware_Lipschitz** (spectral-norm MLP with cost penalty), and (iii) **Lipschitz_NN** (spectral-norm MLP without cost penalty), all with rank targets and walk-forward HP selection. Ridge_Slow remains best (Table 9). CostAware_Linear achieves 7.1% turnover but sacrifices gross signal (net SR 0.37). Lipschitz_NN fails entirely (SR 0.07, 33% turnover): even with slow inputs, the nonlinear prediction surface generates instability. Coupling signal and turnover in a single loss forces a tradeoff that the simple upstream fix avoids.

Long-short portfolios. Short-selling is restricted or prohibitively expensive in most EM. On five markets (BRA, IND, KOR, TWN, THA) at 3× costs plus 50 bps borrowing cost, short-only net SRs are negative for *every* strategy (MomSort −0.44, Ridge −0.76, XGBoost −0.98): the short leg's 19–58% turnover overwhelms any signal, reinforcing long-only as the relevant EM evaluation.

Quarterly rebalancing. Quarterly rebalancing lifts Ridge_All from 0.35 to 0.48 at 11% turnover, matching Ridge_Slow (0.47, 10%)

Table 9: Cost-aware alternatives vs. Ridge_Slow (rank targets). Mean OOS across 15 markets at 3× costs. No cost-aware variant matches the upstream fix.

Strategy	Net SR	Turnover	Gross SR
Ridge_Slow	**0.47**	10.2%	0.61
CostAware_Linear	0.37	7.1%	0.47
MomentumSort	0.36	17.0%	0.58
CostAware_Lipschitz	0.34	12.6%	0.52
Lipschitz_NN	0.07	33.4%	0.48

with no feature selection at all; Ridge_Slow_Q reaches 0.49 at 7% and MomentumSort_Q barely changes (0.37). This is consistent with the ceiling: trading a third as often shrinks turnover regardless of signal mix, so the ceiling does not depend on the mechanism. The two routes are performance-equivalent in our sample but differ operationally: slow-feature selection retains monthly reactivity and an ex-ante diagnostic, while quarterly rebalancing holds stale fast-feature positions for three months and is validated only ex post. Stacking both adds little (0.49), consistent with substitutes on the same frontier.

Statistical significance. Single markets are too short (56–93 OOS months) to separate strategies. Jobson-Korkie tests with the Memmel [12] correction reject at 5% in only one market, for MomSort over LightGBM. Pooled across markets, the sort's edge over the trees is significant (Wilcoxon $p = 0.008$ for XGBoost, $p = 0.003$ for LightGBM), and its edge over Ridge is not ($p = 0.45$).

Ridge_Slow's advantage over MomSort is consistent but too small to test reliably. A block bootstrap of pooled monthly net returns (10,000 replications, resampling dates to keep cross-market dependence) gives one-sided $p = 0.10$ for the annualized SR difference, with 95% CI $[-0.08,\ 0.51]$. Ridge_Slow comes out ahead in 90% of samples, and a t-test on pooled monthly differences agrees ($p = 0.22$). At this effect size ($d = 0.08$), 80% power at the 5% level would take ∼980 months of OOS data.

The frontier claim can be tested jointly. Across the eight interventions, the per-market SR gain over MomentumSort averages +0.11. An exact sign-flip test over markets (2^{15} patterns) gives $p = 0.008$ (Wilcoxon $p = 0.015$). The eight variants share mechanisms and the markets are correlated, so we count this as one piece of joint evidence.

Universe and weighting. The ranking holds without the smallest universe (GRC, $N = 50$): Ridge_Slow 0.45 vs. MomSort 0.36, with the ceiling band at 0.42–0.47. It also holds with value weighting: Ridge_Slow 0.40, MomSort 0.31, and Ridge_All 0.30, with Ridge_Slow beating MomSort in 10/15 markets and positive in 14/15.

7 Conclusion

ML equity strategies in emerging markets face a cost-speed ceiling. Across 15 markets, eight intervention variants reach net Sharpe ratios of ∼0.45–0.49 at 3× EM transaction costs. A linear frontier (Equation 2) traces the ceiling: at each cost level, the ex-ante Feature Speed Index explains 64–65% of cross-sectional SR degradation and 73% of realized turnover.

Which intervention a strategy uses matters less than whether it cuts the turnover that fast features cause. Feature selection, prediction smoothing, and less frequent rebalancing all reach the same ceiling; restricting Ridge to slow features is the simplest route (SR 0.47, 10% turnover). Model smoothness is a second constraint: piecewise-constant tree surfaces generate turnover even on stable inputs, and smoothed trees still fall short of the ceiling. The ceiling supports Gârleanu and Pedersen's [7] prediction that optimal portfolios overweight slow signals, and the frontier can be computed before any capital is traded.

Code, result files, and a script that checks the reported numbers are available at https://github.com/24GUNV/feature-speed-release.

Acknowledgments

We thank Tutanon Sinthuprasith, Head of SCBX R&D, for leadership and support of this work.

References

[1] Vitor Azevedo, Christopher Hoegner, and Mihail Velikov. 2024. The Expected Returns on Machine-Learning Strategies. *Journal of Finance* (2024). Forthcoming.
[2] Andrew Y Chen, Matthias X Hanauer, and Tobias Kalsbach. 2025. Non-Standard Errors in Machine Learning: Evidence from Asset Pricing. *Working Paper* (2025).
[3] Shane A Corwin and Paul Schultz. 2012. A Simple Way to Estimate Bid-Ask Spreads from Daily High and Low Prices. *The Journal of Finance* 67, 2 (2012), 719–760.
[4] Victor DeMiguel, Alberto Martin-Utrera, Francisco J Nogales, and Raman Uppal. 2020. A Transaction-Cost Perspective on the Multitude of Firm Characteristics. *The Review of Financial Studies* 33, 5 (2020), 2180–2222.
[5] Mikheil Esakia and Felix Goltz. 2025. Machine Learning for Long-Only Portfolio Construction. *Working Paper* (2025).
[6] Andrea Frazzini, Ronen Israel, and Tobias J Moskowitz. 2018. Trading Costs. *Working Paper* (2018).
[7] Nicolae Gârleanu and Lasse Heje Pedersen. 2013. Dynamic Trading with Predictable Returns and Transaction Costs. *The Journal of Finance* 68, 6 (2013), 2309–2340.
[8] Shihao Gu, Bryan Kelly, and Dacheng Xiu. 2020. Empirical Asset Pricing via Machine Learning. *The Review of Financial Studies* 33, 5 (2020), 2223–2273.
[9] Matthias X Hanauer and Tobias Kalsbach. 2023. Machine Learning and the Cross-Section of Emerging Market Stock Returns. *Emerging Markets Review* 55 (2023), 101022.
[10] Theis Ingerslev Jensen, Bryan Kelly, Semyon Malamud, and Lasse Heje Pedersen. 2024. Machine Learning and the Implementable Efficient Frontier. *The Review of Financial Studies* (2024). Forthcoming.
[11] Theis Ingerslev Jensen, Bryan Kelly, and Lasse Heje Pedersen. 2023. Is There a Replication Crisis in Finance? *The Journal of Finance* 78, 5 (2023), 2465–2518.
[12] Christoph Memmel. 2003. Performance Hypothesis Testing with the Sharpe Ratio. *Finance Letters* 1, 1 (2003), 21–23.
[13] Robert Novy-Marx and Mihail Velikov. 2016. A Taxonomy of Anomalies and Their Trading Costs. *The Review of Financial Studies* 29, 1 (2016), 104–147.
[14] Petar Stankov et al. 2024. Factor Investing in Emerging Markets: Turnover-Aware Approaches. *Working Paper* (2024).